Lab 8 - Edge Detection

By: Dr. Irfan

Mohammed Naif Alhadi - 2240007563 - 7MA2

In [1]:
```
import cv2
import numpy as np
import matplotlib.pyplot as plt

imgFile = 'kid1.jpg'
```

Sobel Edge Detector

In [2]:
```
# 1. Load the real image
img = cv2.imread(imgFile)

if img is None:
    raise FileNotFoundError(f"Could not load image: {imgFile}")

# 2. Convert to Grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# 3. Try Original Image, Grayscale, and Gaussian Blur
inputs = [
    ("Original Image", img),
    ("Grayscale", gray),
    ("Gaussian Blur", cv2.GaussianBlur(gray, (5, 5), 0))
]

plt.figure(figsize=(15, 8))

for i, (title, blurred) in enumerate(inputs):
    # 4. Apply Sobel filter
    sobelx = cv2.Sobel(blurred, cv2.CV_64F, 1, 0, ksize=3)
    sobely = cv2.Sobel(blurred, cv2.CV_64F, 0, 1, ksize=3)

    # 5. Calculate Magnitude (Final Edge Map)
    sobel_combined = cv2.magnitude(sobelx, sobely)

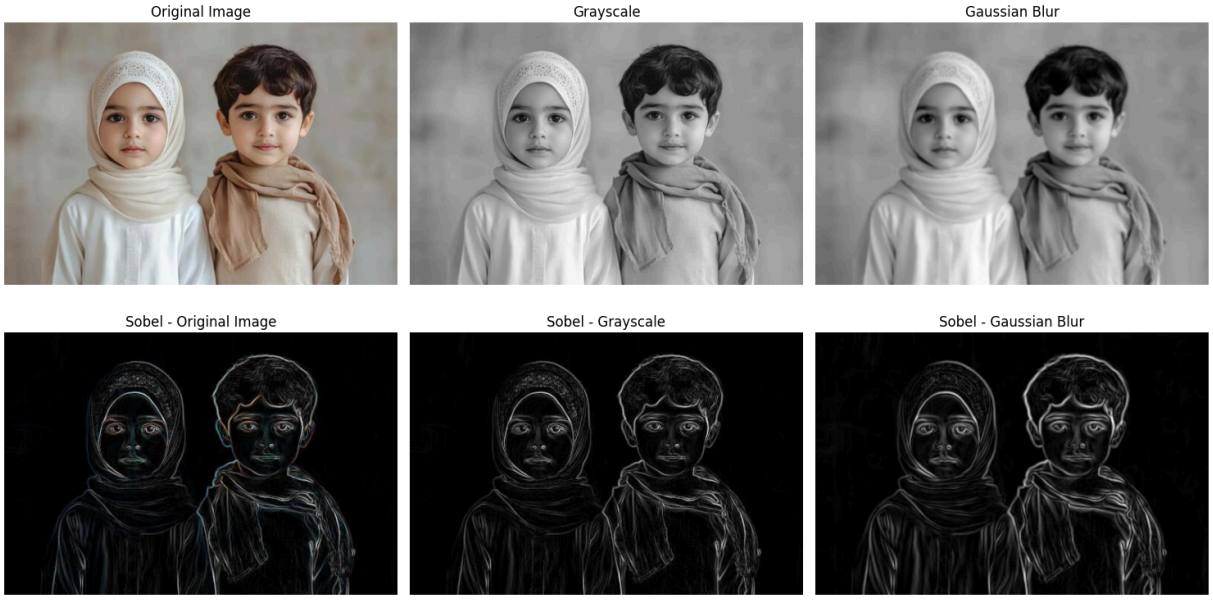
    # 6. Normalize to 0-255 for display
    sobel_final = cv2.normalize(
        sobel_combined, None, 0, 255, cv2.NORM_MINMAX
    ).astype(np.uint8)

    # Visualization: Input
    plt.subplot(2, 3, i + 1)
    if blurred.ndim == 3:
        plt.imshow(cv2.cvtColor(blurred, cv2.COLOR_BGR2RGB))
```

```
    else:
        plt.imshow(blurred, cmap='gray')
    plt.title(title)
    plt.axis('off')

    # Visualization: Sobel Result
    plt.subplot(2, 3, i + 4)
    if sobel_final.ndim == 3:
        plt.imshow(cv2.cvtColor(sobel_final, cv2.COLOR_BGR2RGB))
    else:
        plt.imshow(sobel_final, cmap='gray')
    plt.title(f'Sobel - {title}')
    plt.axis('off')

plt.tight_layout()
plt.show()
```



Robert Edge Detector

Comparing to Sobel

Roberts: Kernel Size --> 2x2 Sobel: Kernel Size --> 3x3

Roberts: Speed --> Faster Sobel: Speed --> Slower

Roberts: Noise Resistance --> Poor (High Senstivity) Sobel: Noise Resistnace --> Good (Built-in Smoothing)

Roberts: Edge Width --> Very thin/sharp Sobel: Edge Width --> Thicker

In [3]:
```
# 1. Load a real image in grayscale
image = cv2.imread(imgFile, 0) #cv2.IMREAD_GRAYSCALE)

# 2. Pre-process: Always blur for Roberts!
```

```
# Its 2x2 kernel is very sensitive to noise.
blurred = cv2.GaussianBlur(image, (5, 5), 0)

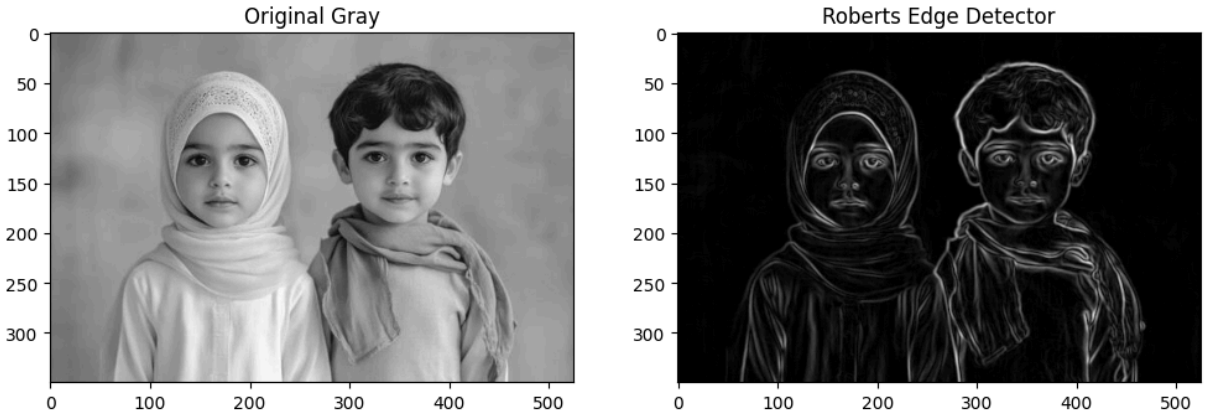
# 3. Define Roberts Cross kernels
kernel_x = np.array([[1, 0], [0, -1]], dtype=np.float64)
kernel_y = np.array([[0, 1], [-1, 0]], dtype=np.float64)

# 4. Apply kernels using convolution
roberts_x = cv2.filter2D(blurred, cv2.CV_64F, kernel_x)
roberts_y = cv2.filter2D(blurred, cv2.CV_64F, kernel_y)

# 5. Calculate Magnitude (Combine X and Y)
# Magnitude = sqrt(Gx^2 + Gy^2)
roberts_combined = cv2.magnitude(roberts_x, roberts_y)

# 6. Convert to uint8 for display
roberts_final = np.uint8(np.absolute(roberts_combined))

# Display
plt.figure(figsize=(12, 6))
plt.subplot(1, 2, 1), plt.imshow(image, cmap='gray'), plt.title('Original Gray')
plt.subplot(1, 2, 2), plt.imshow(roberts_final, cmap='gray'), plt.title('Roberts Ed
plt.show()
```



Prewitt Edge Detector

In [4]:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# 1. Load a real image in grayscale
img = cv2.imread(imgFile)

# 2. Grayscale & Blur (Essential for real photos to reduce grain)
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
blurred = cv2.GaussianBlur(gray, (5, 5), 0)

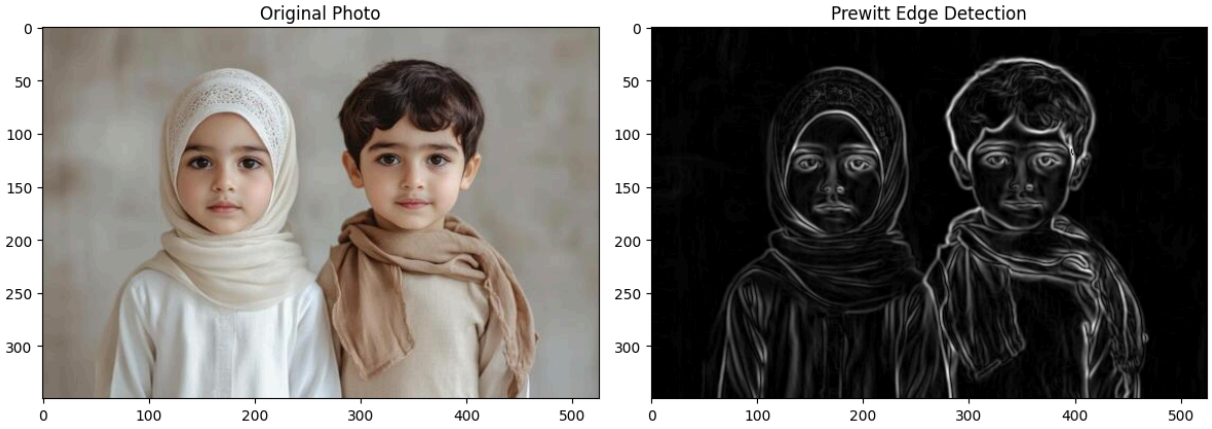
# 3. Define Prewitt Kernels manually
kernel_x = np.array([[-1, 0, 1], [-1, 0, 1], [-1, 0, 1]], dtype=np.float64)
kernel_y = np.array([[-1, -1, -1], [0, 0, 0], [1, 1, 1]], dtype=np.float64)
```

```
# 4. Apply filters
prewitt_x = cv2.filter2D(blurred, cv2.CV_64F, kernel_x)
prewitt_y = cv2.filter2D(blurred, cv2.CV_64F, kernel_y)

# 5. Calculate Magnitude (Combine X and Y)
prewitt_combined = cv2.magnitude(prewitt_x, prewitt_y)

# 6. Normalize to 8-bit for display
prewitt_final = np.uint8(np.absolute(prewitt_combined))

# Visualizing results
plt.figure(figsize=(12, 6))
plt.subplot(1, 2, 1), plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB)), plt.title('
plt.subplot(1, 2, 2), plt.imshow(prewitt_final, cmap='gray'), plt.title('Prewitt Ed
plt.tight_layout()
plt.show()
```



Canny Edge Detector

In [5]:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# 1. Load the real image
img = cv2.imread(imgFile)

# 2. Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# 3. Apply Canny Edge Detection
# Try threshold ratios of 1:2 and 1:3
canny_edges_2 = cv2.Canny(gray, 100, 200)
canny_edges_3 = cv2.Canny(gray, 100, 300)

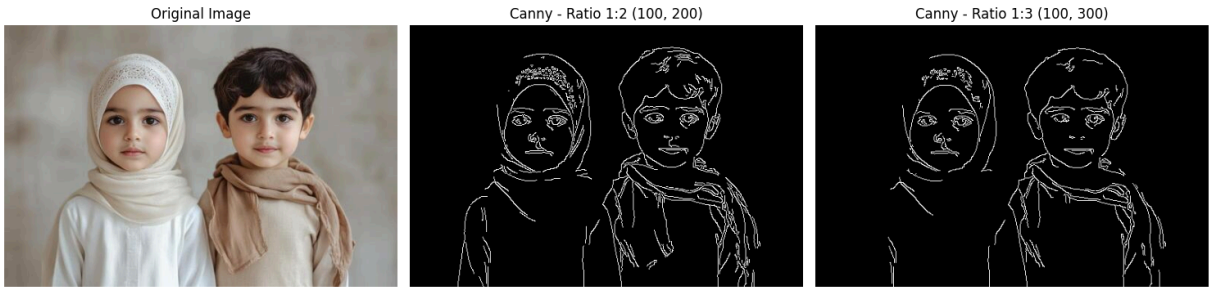
# 4. Visualization
plt.figure(figsize=(15, 5))

plt.subplot(1, 3, 1)
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title('Original Image')
plt.axis('off')
```

```
plt.subplot(1, 3, 2)
plt.imshow(canny_edges_2, cmap='gray')
plt.title('Canny - Ratio 1:2 (100, 200)')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(canny_edges_3, cmap='gray')
plt.title('Canny - Ratio 1:3 (100, 300)')
plt.axis('off')

plt.tight_layout()
plt.show()
```



Multi-Channel Canny Edge Detector

In [6]:
```
def best_color_edge_detector(image_path):
    # 1. Load the color image
    img = cv2.imread(image_path)
    if img is None:
        return "Image not found."

    # 2. Split into B, G, R channels
    channels = cv2.split(img)
    edge_maps = []

    for channel in channels:
        # 3. Pre-process each channel with Gaussian Blur to reduce noise
        blurred = cv2.GaussianBlur(channel, (5, 5), 0)

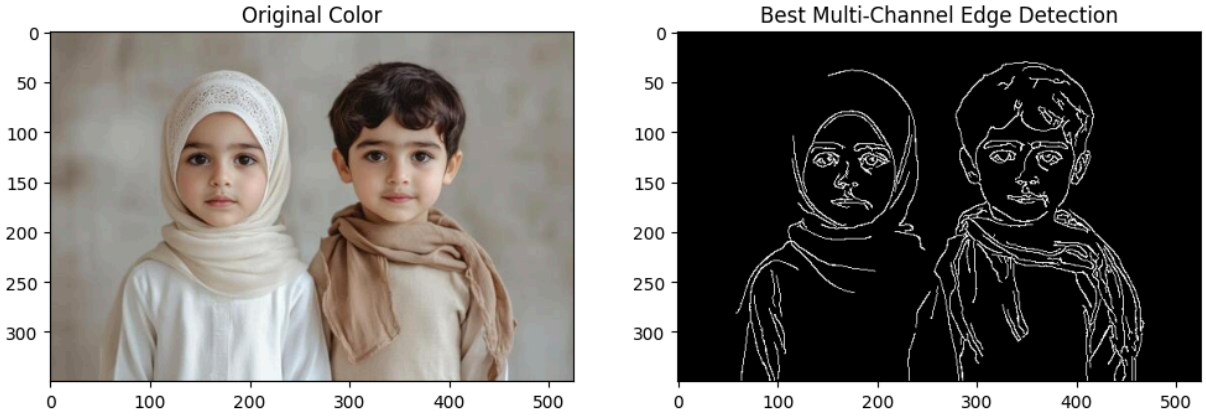
        # 4. Apply Canny to each channel
        # 50 and 150 are the low and high thresholds
        edges = cv2.Canny(blurred, 50, 150)
        edge_maps.append(edges)

    # 5. Merge edges from all channels using bitwise OR
    # This keeps any edge found in ANY of the three color layers
    combined_edges = cv2.bitwise_or(edge_maps[0], edge_maps[1])
    combined_edges = cv2.bitwise_or(combined_edges, edge_maps[2])

    return img, combined_edges

# Run the detector
original, edges = best_color_edge_detector(imgFile)
```

```
# Display results
plt.figure(figsize=(12, 6))
plt.subplot(1, 2, 1), plt.imshow(cv2.cvtColor(original, cv2.COLOR_BGR2RGB)), plt.ti
plt.subplot(1, 2, 2), plt.imshow(edges, cmap='gray'), plt.title('Best Multi-Channel
plt.show()
```



LoG Filter

In [7]:

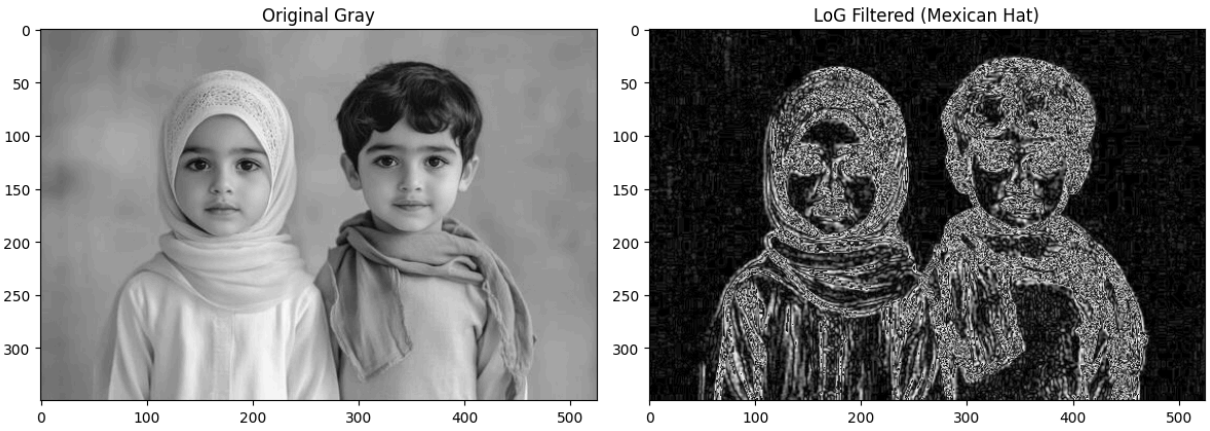
```
# 1. Load a real-world photo in grayscale
img = cv2.imread(imgFile, cv2.IMREAD_GRAYSCALE)

# 2. Step 1: Gaussian Blur (Smoothing)
# Larger kernel/sigma reduces more noise but may miss small edges
blurred = cv2.GaussianBlur(img, (5, 5), 0)

# 3. Step 2: Laplacian Operator (Second Derivative)
# cv2.CV_64F is used to store negative values (crucial for zero-crossings)
laplacian = cv2.Laplacian(blurred, cv2.CV_64F, ksize=5)

# 4. Post-processing: Normalize for display
# Since the Laplacian has negative and positive values, we take the absolute value
log_final = np.uint8(np.absolute(laplacian))

# Visualization
plt.figure(figsize=(12, 6))
plt.subplot(1, 2, 1), plt.imshow(img, cmap='gray'), plt.title('Original Gray')
plt.subplot(1, 2, 2), plt.imshow(log_final, cmap='gray'), plt.title('LoG Filtered (
plt.tight_layout()
plt.show()
```



DoG Filter

In [8]:
```
# 1. Load the real-world photo
img = cv2.imread(imgFile, cv2.IMREAD_GRAYSCALE)

# 2. Apply two Gaussian blurs with different standard deviations
# Low sigma preserves detail; High sigma captures broad structure
blur_narrow = cv2.GaussianBlur(img, (0, 0), sigmaX=1.0)
blur_wide = cv2.GaussianBlur(img, (0, 0), sigmaX=2.0)

# 3. Calculate the Difference (DoG)
# Use float32 to preserve negative values during subtraction
dog = blur_narrow.astype(np.float32) - blur_wide.astype(np.float32)

# 4. Normalize or Threshold for visibility
# Option A: Map to 0-255 range
dog_normalized = cv2.normalize(
    dog, None, 0, 255, cv2.NORM_MINMAX
).astype(np.uint8)

# Option B: Threshold the normalized response
_, dog_binary = cv2.threshold(
    dog_normalized, 127, 255, cv2.THRESH_BINARY
)

# Visualization
plt.figure(figsize=(15, 5))

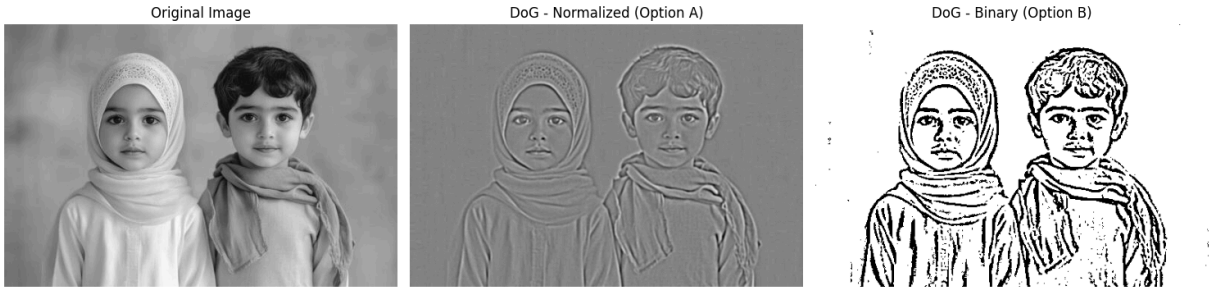
plt.subplot(1, 3, 1)
plt.imshow(img, cmap='gray')
plt.title('Original Image')
plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(dog_normalized, cmap='gray')
plt.title('DoG - Normalized (Option A)')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(dog_binary, cmap='gray')
plt.title('DoG - Binary (Option B)')
```

```
plt.axis('off')

plt.tight_layout()
plt.show()
```



Gradient Edge Detector for Colored Images

In [9]:
```
# 1. Load the real-world color image
image = cv2.imread(imgFile)

# 2. Split the image into its R, G, and B components
# Note: OpenCV uses BGR order by default
b, g, r = cv2.split(image)

def get_channel_gradient(channel):
    # Use Sobel for gradient calculation on a single channel
    # Blur first to reduce noise in that specific color layer
    blurred = cv2.GaussianBlur(channel, (3, 3), 0)
    gx = cv2.Sobel(blurred, cv2.CV_64F, 1, 0, ksize=3)
    gy = cv2.Sobel(blurred, cv2.CV_64F, 0, 1, ksize=3)
    return cv2.magnitude(gx, gy)

# 3. Compute gradients for each color channel
grad_b = get_channel_gradient(b)
grad_g = get_channel_gradient(g)
grad_r = get_channel_gradient(r)

# 4. Combine gradients from all channels
# This ensures edges from color-only transitions are captured
combined_gradient = np.sqrt(grad_b**2 + grad_g**2 + grad_r**2)

# 5. Normalize to 0-255 for visualization
color_edges = cv2.normalize(combined_gradient, None, 0, 255, cv2.NORM_MINMAX).astyp

# Visualization
plt.figure(figsize=(12, 6))
plt.subplot(1, 2, 1), plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB)), plt.title
plt.subplot(1, 2, 2), plt.imshow(color_edges, cmap='inferno'), plt.title('Color-Awa
plt.tight_layout()
plt.show()
```


Original Color Image
Color-Aware Gradient Map
0
50
100
150
200
250
300
0
100
200
300
400
500